

Hugging Face Fine-Tuning, Experiment Tracking & Model Deployment

MLOps_Assignment_2

Roll no. G25AIT2096

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Assignment details: Training and Fine-Tuning BERT for Goodreads Review Classification

Classifying Goodreads Reviews By Book Genre By Maria Antoniak, Melanie Walsh, and the AI for Humanists Team Updated: 2024-11-05

This project demonstrates an end-to-end machine learning workflow including model training, experiment tracking, model versioning, and inference using Hugging Face Transformers, Weights & Biases, and Kaggle.

A BERT model is fine-tuned on Goodreads reviews from the UCSD Book Graph with the goal of predicting the genre of the book being reviewed.

The genres include:

- poetry
- comics & graphic
- fantasy & paranormal
- history & biography
- mystery, thriller, & crime
- romance
- young adult

- This project focused on fine-tuning a BERT-based transformer model for text classification using Goodreads book reviews.
- The objective was to classify reviews according to book genres by leveraging transfer learning from a pretrained language model.
- The workflow included configuring a Kaggle Notebook environment, enabling GPU acceleration, setting up experiment tracking with Weights & Biases (W&B;), and publishing the final trained model to Hugging Face.
- The project emphasized reproducibility, experiment management, and practical MLOps workflow integration in addition to model performance.

Basic steps involved in using BERT and HuggingFace:

1. Divide your data into training and test sets.
 2. Encode your data into a format BERT will understand.
 3. Combine your data and labels into dataset objects.
 4. Load the pre-trained BERT model.
 5. Fine-tune the model using your training data.
 6. Predict new labels and evaluate performance on your test data.
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Why BERT Model –

Architecture & capabilities

- BERT (Bidirectional Encoder Representations from Transformers) is one of the most widely used transformer architectures for NLP tasks.
- Unlike TF-IDF with Logistic Regression, BERT captures contextual meaning from both left and right directions in a sentence.
- Highly effective for understanding semantics of Goodreads reviews, where meaning depends on context, tone, and sentence structure.

Hugging Face integration

- Hugging Face Transformers library provides pretrained checkpoints, tokenizers, and training utilities that simplify fine-tuning.
- Fine-tuning a pretrained BERT model reduces training time significantly compared to training from scratch, while maintaining strong classification performance.

MLOps compatibility

- Kaggle: Cloud-based GPU training
- W&B: Experiment tracking and metric logging
- Hugging Face Hub: Model hosting and inference

Community & support

- Strong community support and extensive documentation available.
- Reduced setup complexity and made debugging easier during implementation.

DistilBERT is 40% smaller and 60% faster than full BERT with only a small accuracy drop. Hence DistilBERT model for this assignment.

Kaggle training setup:

Component	Details
Platform	Kaggle Notebook environment — free GPU access, notebook-based experimentation
Hardware	T4 GPU (enabled via Kaggle settings panel) — used for tokenization and fine-tuning
Libraries & Frameworks	Python Hugging Face Transformers PyTorch Datasets Weights & Biases (W&B)
Secret Management	Kaggle Secrets — API tokens for W&B and Hugging Face stored as notebook secrets (no hardcoding)
Security Practice	MLOps best practice — secure credential handling via environment-level secrets
Workflow Steps	1. Load Goodreads review dataset
	2. Clean and preprocess text data
	3. Tokenize reviews using BERT tokenizer
	4. Fine-tune the pretrained model
	5. Track training metrics with W&B
	6. Save and upload final model to Hugging Face
	7. Create inference script for prediction testing
Reproducibility	Auto-generated requirements.txt and GitHub repository structure for submission
Experiment Tracking	Weights & Biases (W&B) — logs training metrics in real time
Model Storage	Final model uploaded to Hugging Face Hub
Dataset	Goodreads book reviews

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A. Challenges:

Environment & tooling

- Configuring the training environment correctly across Kaggle, Hugging Face, and W&B was one of the most time-consuming challenges.
- Authentication tokens, public/private visibility settings, and package installation errors required troubleshooting.
- Network-related pip installation failures and GPU session resets on Kaggle occasionally interrupted training.

Pipeline complexity

- Understanding transformer-based workflows compared to traditional machine learning pipelines added initial complexity.
- Tokenization, batching, padding, and GPU memory management introduced additional configuration overhead.
- Fine-tuning required careful selection of parameters such as learning rate, batch size, and number of epochs.

B. Learnings:

- How to use a pretrained model as a plug-and-play component without needing to understand its internal architecture.
- How to configure and connect multiple MLOps tools (Kaggle, W&B, Hugging Face) into a unified pipeline.

- How experiment tracking with W&B improves reproducibility and makes training progress visible.
- How to structure an end-to-end MLOps workflow — from data loading to model publishing.
- How to publish and share machine learning artifacts publicly via Hugging Face Hub.
- How GPU sessions and cloud notebook environments (Kaggle) are managed during model training runs.

The Target for this assignment is not to achieve perfect accuracy but rather to understand the workflow of MLOps. Nonetheless,

- More time spent on organising and cleaning the training data before running the pipeline would have made the overall process smoother.
- Logging each training run in W&B with clear labels from the start would have made it easier to track what was tried and what worked.
- Testing the full pipeline on a small sample first — before running the complete training — would have saved time when errors came up midway.

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C. Assignment Links:

Training Platform

Kaggle Notebook (Public Link):

<https://www.kaggle.com/code/g25ait2096/mlops-assignment-2-fine-tuning-classification/>

Hugging Face Model

<https://huggingface.co/g25ait2096/mlops-finetuned-distilbert>

W&B Project Dashboard

https://wandb.ai/g25ait2096-iitj/MLOps-Assign_2

Github Link:

[g25ait2096/MLOps_Assignment_2: Hugging Face Fine-Tuning, Experiment Tracking & Model Deployment](https://github.com/g25ait2096/MLOps_Assignment_2: Hugging Face Fine-Tuning, Experiment Tracking & Model Deployment)

D. Output:

Results

Metric	Value
Accuracy	60.31%
F1 Score	60.31%
Loss	2.358

E. Conclusion

The Target for this assignment was not to achieve perfect accuracy but rather to understand the workflow of MLOps.

This project successfully demonstrated the fine-tuning of a BERT-based model process for Goodreads review classification using Kaggle infrastructure and modern MLOps tools.

The workflow combined model training, experiment tracking, secure configuration management, and reproducible deployment practices.

Beyond achieving classification results, the project highlighted the importance of workflow organization, automation, and collaboration tools in machine learning engineering.
